# The Chaplygin Sleigh for Every Signed Mass Offset: Exact Scattering, Physical Reconstruction, and the Recurrence Boundary

Route-A structural certificate C199

**Abstract**

We give a convention-complete theorem for the classical Chaplygin sleigh with mass $m > 0$, central inertia $J > 0$, and arbitrary signed longitudinal offset $a$. For $a \neq 0$, every non-equilibrium positive-energy orbit is an explicit heteroclinic: its blade rotates through an energy-independent angle and its contact point has two rigorously defined asymptotic lines. We also classify the stable equilibrium half-axis, half-plane Poisson structure, necessarily singular reduced density and its configuration-Haar lift, reversor, and all zero-offset and zero-angular- velocity boundaries. A finite exact/high-precision ledger checks conventions but is not used to prove the continuous theorem. The Poisson form gives only a formal Route-A hint; the absence of an intrinsic prime carrier and target determinant forces rejection.

**Keywords:** Chaplygin sleigh; nonholonomic mechanics; heteroclinic scattering; reconstruction; invariant measure; reversibility.

## 中文摘要

本文对质量、转动惯量均为正且质心纵向偏置可取任意符号的经典 Chaplygin sleigh给出完整定理。偏置非零时，全部非平衡正能轨道均为 显式异宿轨道；刀刃转角与能量无关，接触点具有两条严格定义的渐近直线。本文还统一分类 稳定平衡半轴、半平面泊松结构、必然奇异的不变测度、时间反演以及零偏置等全部边界。 有限精确计算只校验约定，不代替连续参数证明；路线A最终仍被拒绝。
关键词：非完整力学；异宿散射；重构；不变测度；时间反演。

## 1 Convention, equations, and ownership

Let $r = (x, y)$ be the knife contact point, $\theta \in \mathbb{S}^1$ the *blade* angle, $u$ the signed contact velocity along the blade, and $\omega = \dot{\theta}$. Let $a \in \mathbb{R}$ be the signed longitudinal offset from contact point to centre of mass and put $I_c = J + ma^2$. In physical time, the reduced equations and reconstruction are

$$\dot{r} = u\mathbf{e}(\theta), \quad \dot{\theta} = \omega, \quad \dot{u} = a\omega^2, \quad \dot{\omega} = -\frac{ma}{I_c}u\omega, \qquad \mathbf{e}(\theta) = (\cos\theta, \sin\theta). \tag{1}$$

They imply

$$H = \frac{m}{2}u^2 + \frac{I_c}{2}\omega^2 = \text{constant}. \tag{2}$$

The model, its nonholonomic reduction and Hamiltonization context are classical [1, 2, 3]. We claim a signed-parameter synthesis and certificate, not priority for those results.

## 2 All nonzero-offset scattering orbits

**Theorem 1** (signed-offset normal form). *Assume $a \neq 0$, $H > 0$, and $\omega(0) \neq 0$. Define*

$$R = \sqrt{2H/m}, \quad A = \frac{m|a|R}{I_c}, \quad \eta = \frac{\sqrt{I_c/m}}{|a|}, \quad \sigma = \operatorname{sgn}\omega(0).$$

*There are unique $t_0 \in \mathbb{R}$ and $\theta_0$ modulo $2\pi$ such that, with $s = A(t - t_0)$,*

$$u = \operatorname{sgn}(a)R\tanh s, \quad \omega = \sigma R\sqrt{m/I_c}\,\operatorname{sech} s, \quad \theta = \theta_0 + \sigma\eta\arcsin(\tanh s). \tag{3}$$

*It joins $(-\operatorname{sgn}(a)R, 0)$ to $(\operatorname{sgn}(a)R, 0)$ and its blade-angle deflection is*

$$\Delta\theta = \theta_+ - \theta_- = \sigma\pi\eta, \tag{4}$$

*which is independent of $H$.*

*Proof.* The sign of a nonzero $\omega$ cannot change by uniqueness. On the energy ellipse write $u = \operatorname{sgn}(a)R\tanh s$ and $\omega = \sigma R\sqrt{m/I_c}\operatorname{sech} s$. The first reduced equation gives $\dot{s} = A$; substitution gives the second. Since $d\{\arcsin(\tanh s)\}/ds = \operatorname{sech} s$, integration of $\dot{\theta} = \omega$ proves (3). Taking $s \to \pm\infty$ proves both endpoints and (4). ☐

The endpoint table makes the signed convention explicit:

| | $t \to -\infty$ | $t \to +\infty$ |
|---|---|---|
| $u$ | $-\operatorname{sgn}(a)R$ | $\operatorname{sgn}(a)R$ |
| $\omega$ | $0^\sigma$ | $0^\sigma$ |
| $\theta$ | $\theta_0 - \sigma\eta\pi/2$ | $\theta_0 + \sigma\eta\pi/2$ |

**Revision focus.** Round 0 freezes the signed convention and proves the explicit all-energy heteroclinic scattering core.

## Declarations

**Data and code availability.** Package-local synthetic exact evidence and deterministic code accompany the manuscript; no observational data are used.

**Ethics.** No human, animal, personal or sensitive data are used; approval is not applicable.

**Author contributions (CRediT).** This anonymous certificate records Conceptualization, Formal analysis, Software, Validation, Writing—original draft, and Writing—review and editing. AI systems are not authors.

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**Conflicts of interest.** None known.

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## References

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